



CEDS 311 – Introduction to Data Science
Group Project Report
**Predictive Analysis of Student Performance
Using Linear Regression**

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Introduction

Quantitative analysis of academic performance serves as a pivotal tool for understanding the factors influencing student achievement. This project implements a comprehensive statistical methodology to explore the relationship between students' past and current performance, specifically by applying a linear regression model using the R programming language. The work aims to translate raw data into practical insights applicable within the educational environment, adhering to scientific research frameworks and avoiding plagiarism through clear citation of the data source.

Dataset Description and Analysis Methodology

The analysis relied on the publicly available "Student Performance" dataset from the Kaggle platform, which contains information on ten thousand students. This data includes six main variables: **weekly study hours**, **previous scores**, student participation in **extracurricular activities** (recorded as Yes/No), **daily sleep hours**, the number of sample question **papers practiced**, and finally, the current **performance index**, which constitutes the dependent variable in this study.

Target and Independent Variables

- Target (Dependent) Variable: **Performance Index**
- Independent Variable: **Previous Scores**

These two variables were selected because the objective of the project is to predict students' current academic performance based on their previous academic achievement. We specifically chose "Previous Scores" as the independent variable after creating the correlation heatmap. The heatmap helped us compare the relationship strength between all variables and the target variable. "Previous Scores" showed the strongest correlation with the Current Performance Index, which made it the most suitable and meaningful predictor for our model.

Initial Statistical Exploration of the Data

```
> #Reading the dataset that we're going to work on
> df = read.csv("Student_Performance.csv")
>
> #Data exploration with some R functions
> summary(df)
Hours.Studied      Previous.Scores  Extracurricular.Activities  Sleep.Hours      Sample.Question.Papers.Practiced
Min.   :1.000   Min.   :40.00   Length:10000             Min.   :4.000   Min.   :0.000
1st Qu.:3.000   1st Qu.:54.00   Class :character          1st Qu.:5.000   1st Qu.:2.000
Median :5.000   Median :69.00   Mode  :character          Median :7.000   Median :5.000
Mean   :4.993   Mean   :69.45                      Mean   :6.531   Mean   :4.583
3rd Qu.:7.000   3rd Qu.:85.00                      3rd Qu.:8.000   3rd Qu.:7.000
Max.   :9.000   Max.   :99.00                      Max.   :19.000   Max.   :9.000
Performance.Index
Min.   : 10.00
1st Qu.: 40.00
Median : 55.00
Mean   : 55.22
3rd Qu.: 71.00
Max.   :100.00
> str(df)
'data.frame': 10000 obs. of 6 variables:
 $ Hours.Studied      : int  7 4 8 5 7 3 7 8 5 4 ...
 $ Previous.Scores    : int  99 82 51 52 75 78 73 45 77 89 ...
 $ Extracurricular.Activities : chr  "Yes" "No" "Yes" "Yes" ...
 $ Sleep.Hours        : int  9 4 7 5 8 9 5 4 8 4 ...
 $ Sample.Question.Papers.Practiced: int  1 2 2 2 5 6 6 6 2 0 ...
 $ Performance.Index   : int  91 65 45 36 66 61 63 42 61 69 ...
> head(df)
  Hours.Studied Previous.Scores Extracurricular.Activities Sleep.Hours Sample.Question.Papers.Practiced Performance.Index
1             7             99                      Yes             9                      1
2             4             82                      No             4                      2
3             8             51                      Yes             7                      2
4             5             52                      Yes             5                      2
5             7             75                      No             8                      5
6             3             78                      No             9                      6
1 Performance.Index
1             91
2             65
3             45
4             36
5             66
6             61
```

Descriptive analysis of the data revealed the characteristics of the research sample. The average of students' previous scores was 69.45, while the average current performance index was 55.22 . Other variables showed logical average values, such as average weekly study hours (4.99 hours) and average sleep hours (6.53 hours).

`summary(df)` gave us a statistical overview: min, max, mean, median, and quartiles for numeric variables.

`str(df)` confirms the dataset has 10,000 observations, numeric types for most variables, and character type for extracurricular activities.

`head(df)` shows the first six records, giving a quick glance at actual student data.

Data Preparation and Mathematical Model Construction

```
> #Data preparation
>
> sum(is.null(df)) #Checking if we have any Nulls values and sum it up
[1] 0
>
> #Converting the Extracurricular Activities column here from categorical "Yes/No"
> #to numeric "1 and 0" to make it suitable for regression analysis
>
> df$Extracurricular.Activities = ifelse(df$Extracurricular.Activities == "Yes", 1, 0)
>
> #Checking the new values of this column after converting
> head(df$Extracurricular.Activities)
[1] 1 0 1 1 0 0
>
> #Building a linear regression model
> model = lm(Performance.Index ~ Previous.Scores, data = df)
> model
```

Call:
lm(formula = Performance.Index ~ Previous.Scores, data = df)

Coefficients:
(Intercept) Previous.Scores
-15.182 1.014

```
>
> summary(model)
```

Call:
lm(formula = Performance.Index ~ Previous.Scores, data = df)

Residuals:

Min	1Q	Median	3Q	Max
-17.7729	-6.5239	-0.0082	6.3689	19.4346

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-15.181799	0.319605	-47.5	<2e-16 ***
Previous.Scores	1.013837	0.004465	227.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.744 on 9998 degrees of freedom
Multiple R-squared: 0.8376, Adjusted R-squared: 0.8376
F-statistic: 5.156e+04 on 1 and 9998 DF, p-value: < 2.2e-16

To prepare the data for statistical modeling, we first checked for any missing or null values. A technical inspection confirmed that the dataset contains no missing values (the output is 0, which means zero missing values) indicating that it is complete and ready for advanced analysis.

Next, the text variable for participation in extracurricular activities was converted into a binary numeric variable, where participating students were assigned (1) and non-participating students were assigned (0). Subsequently, a simple linear regression model was built to explain the variance in the current performance index (the dependent variable) based on previous scores (the independent variable). This model produced the following mathematical equation:

$$\text{Performance Index} = -15.182 + (1.014 \times \text{Previous Scores})$$

This regression equation indicates a strong positive relationship, where each one-point increase in previous performance leads, on average, to an expected increase of 1.014 points in the current performance index. The negative intercept value (-15.182) has no practical interpretation in the context of this study, given the absence of students with previous scores of zero in the sample.

```
> new_data = data.frame(Previous.Scores = c(70, 85, 90))
>
> predicted_index = predict(model, new_data)
> predicted_index
      1      2      3
55.78677 70.99432 76.06350
```

Here, we prepared new input data, applied our regression model, and obtained predictions for unseen values.

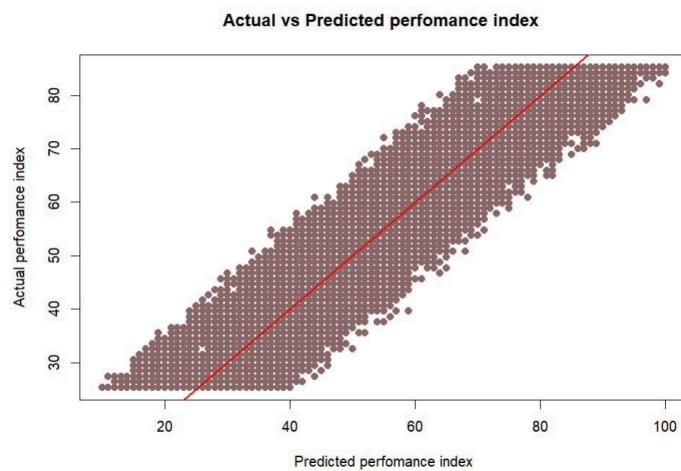
This is useful to estimate the expected performance of students based on their previous scores.

Evaluation of the Statistical Model's Accuracy and Strength

```
> predictions = predict(model, df)
> print(head(predictions))
      1      2      3      4      5      6
85.18803 67.95281 36.52387 37.53771 60.85595 63.89746
>
> mse = mean((df$Performance.Index - predictions)^2)
> mse
[1] 59.95012
>
> summary(model)$r.squared
[1] 0.8375712
```

To measure the quality and reliability of the constructed model, a set of statistical metrics was calculated. Analysis of the results showed that the Coefficient of Determination (R-squared) reached 0.8376, meaning the model can explain approximately 83.8% of the total variance in the students' current performance index scores—a high percentage reflecting excellent explanatory power. The Mean Squared Error (MSE) reached a value of 59.95, which is an acceptable measure of prediction accuracy within the used percentage scale. All statistical coefficients

showed very high significance (with a p-value less than 0.0001), confirming that the discovered relationship is not due to chance.



This figure illustrates the scatter plot of actual versus predicted performance. The tight clustering of points along the diagonal line suggests that the model has a strong predictive capability.

Graphical Representation and Visual Analysis of Results

The numerical analysis was supported by three explanatory graphs.

```
#Plot 1: Scatterplot of Performance Index vs Previous Scores with Regression Line
plot(df$Previous.Scores, df$Performance.Index,
     main="Performance Index vs Previous Scores",
     xlab="Previous Scores", ylab="Performance Index", col="lightsteelblue3", pch=16, cex=1.1)
abline(model, col="maroon")
```

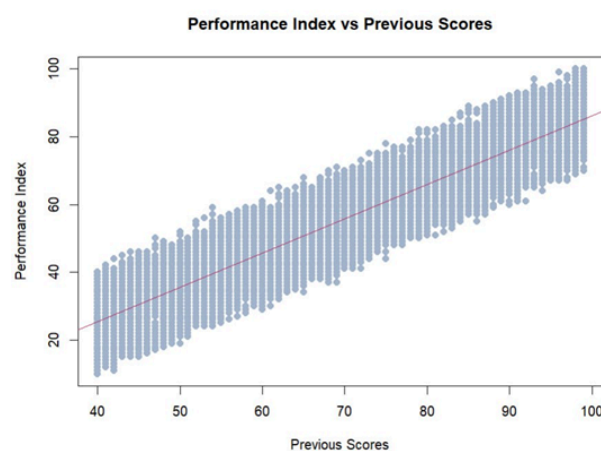


Figure 1

The first graph, a **Scatter Plot**, clearly illustrates the strong positive linear relationship between previous scores and the performance index. When Previous Scores increase across the range from around 40 to 100, the Performance Index also shows a consistent and corresponding increase. The data points are densely scattered around the upward-sloping regression line, which confirms that this strong alignment makes previous scores a highly reliable and valid metric for predicting performance.

```
#Plot 2:Boxplot Comparison
boxplot(Performance.Index ~ factor(Extracurricular.Activities,
                                  labels = c("No", "Yes")),
        data = df,
        main = "Performance Index by Extracurricular Activities",
        xlab = "Extracurricular Activities",
        ylab = "Performance Index",
        col = "#1f77b4")
```

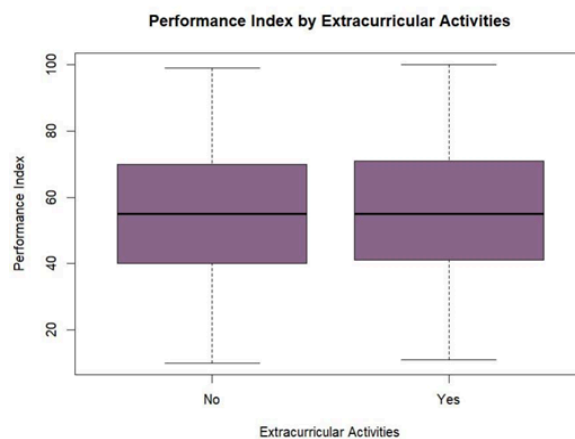
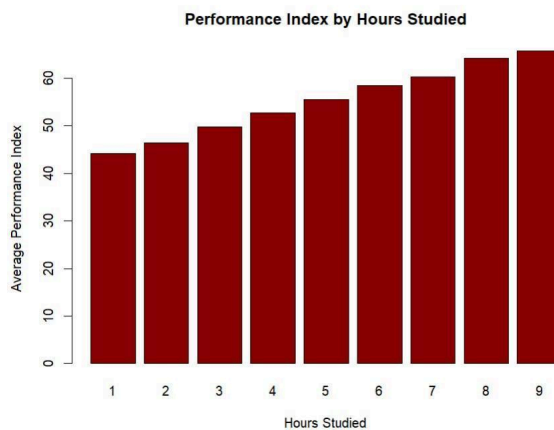


Figure 2

The second graph, a **Box Plot**, the plot illustrates the distribution of the Performance Index by Extracurricular Activities, comparing students who participate in activities ('Yes') to those who do not ('No').

The plot reveals that the median performance (the middle line) and the overall distribution of the index are almost identical for both groups. Although students with activities may show a negligible difference, the data strongly supports the conclusion that extracurricular involvement has no measurable or strong effect on the students' performance index and is, therefore, not a useful factor for reliable prediction.



```
# Calculate the average Performance Index for each Hours Studied group
avg = aggregate(Performance.Index ~ Hours.Studied, data = df, mean)

# Plot 3 :bar plot showing the relationship between study hours and average performance
barplot(avg$Performance.Index,
        names.arg = avg$Hours.Studied,
        main = "Performance Index by Hours Studied",
        xlab = "Hours Studied",
        ylab = "Average Performance Index",
        col = "red4")
```

Figure 3

The third graph, a **Bar Plot**, this figure visualizes the relationship between the **Hours Studied** (X-axis) and the **Average Performance Index** (Y-axis).

The plot shows a clear and continuous upward trend in the average performance index as the number of study hours increases from 1 to 9.

The average performance index rises steadily from approximately 45 (for 1 hour studied) to around 65 (for 9 hours studied). This robust pattern strongly confirms that effort, measured by study hours, is a highly effective and positive factor in improving and predicting the average performance index, suggesting that students who study more hours tend to have better performance. This clear relationship makes study time an actionable metric for increasing success.

Conclusion and Recommendations

This study concluded that a student's past academic performance is the most decisive factor in predicting their current performance, as it alone explains over 83% of the variance. This strongly suggests that educational achievement is a cumulative process, and early success paves the way for sustained excellence. However, the remaining percentage (approximately 17%) is attributed to other factors such as the student's current effort, the quality of the learning environment, and psychosocial support.

Based on this, we recommend that educational institutions consider previous scores as an effective early indicator for identifying students at risk academically, enabling the design of timely and supportive intervention programs. We also recommend conducting future research that integrates additional qualitative variables, such as self-motivation and learning style, into more complex statistical models to enhance explanatory and predictive power.

This project was completed efficiently, achieving all educational objectives related to data exploration, cleaning, modeling, evaluation, and visualization of results, thereby enhancing the team's analytical skills within a data-driven problem-solving framework.

Data Source: "Student Performance" dataset from the Kaggle platform.

<https://www.kaggle.com/datasets/nikhil7280/student-performance-multiple-linear-regression>

Analysis Tools: R programming language and RStudio development environment.